

Murphy Diggins

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PROFESSIONAL SUMMARY

Biomedical Engineering student specializing in biomaterials with hands-on research experience across two university labs, including PDMS substrate fabrication, mammalian and bacterial cell culture, and CAD-driven prototype design. Applies engineering and wet lab skills to device development, materials characterization, and validated experimental workflows in both neuroscience and plant-microbe interaction research contexts.

EDUCATION

University of Wisconsin–Madison

Madison, WI

B.S. Biomedical Engineering, Biomaterials Track. Involvement: Phi Delta Theta (Member, Judicial Board), Biomedical Engineering Society, IFC Softball, Party of Poets, Humorology.
GPA: 3.50/4.00.

Dec. 2027

Relevant Coursework: Biomaterials, Biomechanics, Tissue Engineering, Microscopy, Design & Leadership, Organic Chemistry

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Franck Laboratory, University of Wisconsin–Madison

Madison, WI

- Fabricate PDMS substrates and design experimental fixtures using CAD and 3D printing to support investigation of mechanical strain and substrate stiffness effects on microglial activation, processing approximately 15 samples per week over 12 weeks.
- Identified that microglia exhibit substantially greater activation responses in tri-culture with neurons and astrocytes than in isolated culture, supporting the hypothesis that DAMPs released by damaged cells drive immune response and microglial activation.
- Execute end-to-end microglial isolations including cell feeding and maintenance under sterile conditions using primary cell culture, immunocytochemistry, fluorescence microscopy, and ImageJ/Fiji, sustaining a consistent weekly experimental cadence.

Undergraduate Research Assistant

Mar. 2026 – Aug. 2026

Pfleger Laboratory, University of Wisconsin–Madison

Madison, WI

- Co-designed and led fabrication of polymer-based biomimetic root models using ethyl cellulose and PDMS, benchmarked against flat control surfaces to assess bacterial binding capability.
- Quantified bacterial adhesion via CFU assays using *E. coli* and an *S. stutzeri* strain expressing a binding protein, finding notably greater binding to root mimics than to flat surfaces and establishing the system as a viable tool for testing plant-microbe interactions.
- Performed fluorescence staining and imaging of cell-seeded substrates to characterize surface colonization across material conditions.

Park Ranger, Paraprofessional, Pool Technician, Golf Course Guest Services and Grounds Crew

2021 – 2025

Various Employers

PROJECTS

Wearable Thermal Management Device

Independent Project

- Led design of a sensor-controlled cooling device, independently validating performance at 77°C ambient temperature while maintaining under 15°C surface temperatures.
- Built a Bluetooth mobile app supporting live graphing, three-sensor temperature monitoring, data logging, and historical analysis.

Bone Graft Biomaterials and Bioreactor Design

- Led biomaterial selection for a bone graft system and contributed to bioreactor design and electronics integration.
- Designed a custom sample holder in SolidWorks and manufactured it using manual lathe and mill machining.

SKILLS

Lab: Mammalian & Bacterial Cell Culture, Immunocytochemistry, Sterile Technique, Biological Safety Cabinet (BSC), Fluorescence Microscopy, ImageJ/Fiji, PDMS, CFU Assays, Biofilm Assays

Engineering: SolidWorks, Fusion 360, CAD, 3D Printing, Rapid Prototyping, Lathe & Mill, Experimental Design

Software/Electronics: MATLAB, Arduino IDE, Bluetooth/BLE, Sensor Integration, PreForm